

Role Form Fields & KPI Checklist

All form fields each mobile role submits · database keys · computed values returned

REQ Required

AUTO Auto-captured

COMPUTED System-calculated

OPT Optional

Worker

Field Operative — reports incidents, near misses, and shift attendance

Fields: **0/40**

Incident Report

incidents

- ☐ Incident Date & Time
incident_date_time DateTime **REQ**
- ☐ Location / Working Station
location_station_id FK → working_stations **REQ**
- ☐ Incident Type
incident_type
Injury / Dangerous Occurrence / Property Damage / Environmental
- ☐ Severity
severity Minor / Moderate / Severe / Lost Time / Fatal **REQ**
- ☐ Description
description Text **REQ**
- ☐ Immediate Cause
immediate_cause String(255) **OPT**
- ☐ Number of Persons Involved
number_persons_involved Integer
- ☐ Anyone Injured?
anyone_injured Yes / No **REQ**
- ☐ Injured Person Name (if injured)
injured_person_name String(255)
- ☐ Body Part Injured (if injured)
injured_body_part String(255)
- ☐ Linked Hazard
hazard_id FK → hazards **OPT**
- ☐ Permit Active at Time?
permit_active Yes / No
- ☐ Control Failure?
control_failure Yes / No
- ☐ Hazard Still Present?
hazard_still_present Yes / No
- ☐ Immediate Actions Taken
immediate_actions_taken Text
- ☐ Witnesses
witnesses_json JSON array of names **OPT**
- ☐ Evidence / Photo Upload
evidence_json JSON array (file refs) **OPT**

VALUES WORKER GETS

PERSONAL DASHBOARD

- **My Incidents This Period**
COUNT(incidents WHERE reported_by = me)
- **My Near Misses**
COUNT(near_misses WHERE reported_by = me)
- **Hours Logged (Month)**
SUM(actual_hours_worked WHERE employee = me)
- **My Open CAPA Actions**
CAPA actions assigned to me, status ≠ Completed
- **Incident Status Feed**
workflow_status on my reported incidents

NOTIFICATIONS

- **Supervisor Acknowledged**
acknowledged_at set on incident
- **Investigation Complete**
investigation_status = "Completed"
- **Shift Confirmed**
supervisor_id set on shift record

- ☐ GPS Location
gps_latitude · gps_longitude Decimal **AUTO**
- ☐ Reported By
reported_by FK → employees (session user) **AUTO**

Near Miss Report **near_misses**

- ☐ Event Date & Time
event_date_time DateTime **REQ**
- ☐ Location / Working Station
location_station_id FK → working_stations **REQ**
- ☐ Description
description Text **REQ**
- ☐ Potential Consequence
potential_consequence String(255)
- ☐ Severity
severity Low / Medium / High
- ☐ Linked Hazard
hazard_id FK → hazards **OPT**
- ☐ Underlying Cause
underlying_cause String(255)
- ☐ Control Failure?
control_failure Yes / No
- ☐ Hazard Still Present?
hazard_still_present Yes / No
- ☐ Witnesses
witnesses_json JSON array **OPT**
- ☐ Evidence / Photos
evidence_json JSON array **OPT**
- ☐ GPS Location
gps_latitude · gps_longitude **AUTO**
- ☐ Reported By
reported_by FK → employees **AUTO**
- ☐ CAPA Escalation?
capa_escalation Yes / No

Shift Check-In **shift_schedules**

- ☐ Shift Date
shift_date Date **REQ**
- ☐ Shift Type
shift_type Morning / Afternoon / Night
- ☐ Shift Start Time
shift_start Time
- ☐ Shift End Time
shift_end Time
- ☐ Actual Hours Worked
actual_hours_worked Decimal(4,1) **REQ**

☐

Working Station

station_id FK → working_stations

☐

Employee (self)

employee_id FK → employees

AUTO

Supervisor

Site Supervisor — permits, investigations, CAPA updates, safety walks, shift management

Fields: 0/44

Permit-to-Work permits_to_work

- ☐ Permit Type
permit_type_id
FK → permit_types (Hot Work / Cold Work / Electrical / Confined Space / LOTO...)
REQ
- ☐ Date Issued
date_issued Date REQ
- ☐ Time Issued
time_issued Time
- ☐ Location / Station
location_station_id FK → working_stations
- ☐ Work Description
work_description Text REQ
- ☐ Duration Requested (hrs)
duration_requested_hours Integer
- ☐ Number of Workers
number_of_workers Integer
- ☐ Validity Start
validity_start DateTime
- ☐ Validity End
validity_end DateTime
- ☐ Work Start Actual
work_start_actual DateTime OPT
- ☐ Work End Actual
work_end_actual DateTime OPT
- ☐ Status
status Active / Closed / Expired REQ
- ☐ Deviation Reported?
deviation_reported Yes / No
- ☐ Incident Occurred?
incident_occurred Yes / No
- ☐ Supervisor Notes
supervisor_notes Text OPT
- ☐ Issued By (self)
issued_by FK → employees AUTO

Incident Investigation incidents (update)

- ☐ Investigation Status
investigation_status In Progress / Completed REQ
- ☐ Root Cause
root_cause String(255)
- ☐ Root Cause Category
root_cause_category Human / Equipment / Environment / Process

VALUES SUPERVISOR GETS

ZONE KPIS

TRIR

Injury Rate (Zone)
(Injuries × 200,000) ÷ Man-hrs
incidents + shift_schedules (my area)

NMR

Near Miss Ratio
Near Misses ÷ Recordable Injuries
near_misses / incidents (my zone)

PTW

Open Permits Under Me
COUNT(status = Active, issued_by = me)
permits_to_work

CAPA

Pending CAPA Updates
status IN (Open, In Progress) for my team
capa_actions (responsible_person)

SAFETY WALK METRICS

WLK

Follow-Up Rate
follow_up_required=Yes ÷ Total × 100
safety_walks (my area)

WLK

Avg Compliance Rating
AVG(compliance_rating)
safety_walks

WLK

Critical Issues Count
SUM(critical_issues)
safety_walks (this period)

WORKFORCE

HRS

Total Man-Hours (Team)
SUM(actual_hours_worked)
shift_schedules (my team)

INV

Investigations Queue
investigation_status = null or In Progress
incidents assigned to me

- ☐ 5-Why Analysis
five_why_analysis JSON (structured 5-step form)
- ☐ Immediate Actions Taken
immediate_actions_taken Text
- ☐ Days Away from Work
days_away Integer
- ☐ Severity Classification
severity_classification String(50)
- ☐ CAPA Generated?
capa_generated Yes / No
- ☐ Regulatory Notified?
regulatory_notified Yes / No
- ☐ Supervisor Signature
supervisor_signature String(255) / e-sign ref
- ☐ Escalation Reason (if escalated)
escalation_reason Text OPT

CAPA Action Update

capa_actions (update)

- ☐ Status
status Open / In Progress / Completed / Overdue REQ
- ☐ Effectiveness Rating
effectiveness_rating Integer 1–5 REQ when Completed

Safety Walk Inspection

safety_walks

- ☐ Inspection Date & Time
inspection_date_time DateTime REQ
- ☐ Location / Station
location_station_id FK → working_stations REQ
- ☐ Inspection Type
inspection_type Routine / Compliance / Follow-up
- ☐ Issues Found
issues_found Integer (count)
- ☐ Critical Issues
critical_issues Integer (count)
- ☐ Housekeeping Rating
housekeeping_rating Integer 1–5
- ☐ Compliance Rating
compliance_rating Integer 1–5
- ☐ Follow-Up Required?
follow_up_required Yes / No
- ☐ Inspector (self)
inspector_id FK → employees AUTO

Shift Schedule Management

shift_schedules

- ☐ **Employee**
employee_id FK → employees **REQ**
- ☐ **Shift Date**
shift_date Date **REQ**
- ☐ **Shift Type**
shift_type Morning / Afternoon / Night
- ☐ **Shift Start / End**
shift_start · shift_end Time
- ☐ **Actual Hours Worked**
actual_hours_worked Decimal(4,1)
- ☐ **Station Assignment**
station_id FK → working_stations

CAPA Action Creation capa_actions

- ☐ Linked Incident
incident_id FK → incidents OPT
- ☐ Action Type
action_type Corrective / Preventive / Monitoring REQ
- ☐ Description
description Text REQ
- ☐ Root Cause Addressed
root_cause_addressed String(255)
- ☐ Responsible Person
responsible_person_id FK → employees REQ
- ☐ Due Date
due_date Date REQ
- ☐ Initial Status
status Open (default) AUTO

Incident Close-Out Approval incidents (update)

- ☐ Closure Notes
closure_notes Text REQ
- ☐ Lessons Learned
lessons_learned Text
- ☐ Communicated to Teams?
communicated_to_teams Yes / No
- ☐ Manager Signature
manager_signature String(255) / e-sign ref
- ☐ Workflow Status → Closed
workflow_status closed AUTO on approve
- ☐ Approved At
approved_at DateTime AUTO

Hazard Register Entry hazards

- ☐ Hazard Category
category_id FK → hazard_categories
- ☐ Hazard Name
hazard_name String(255) REQ
- ☐ Severity
severity Low / Medium / High / Critical
- ☐ Probability
probability Rare / Unlikely / Possible / Likely / Almost Certain
- ☐ Description
description Text
- ☐ Location / Station

KPIS SAFETY MANAGER GETS

MODULE 1 — INCIDENTS

TRIR

Total Recordable Injury Rate
(Recordables × 200,000) ÷ Man-hrs
OSHA 300A

LTIFR

LTI Frequency Rate
(LTIs × 1,000,000) ÷ Man-hrs
IOGP / ISO 45001

LTISR

LTI Severity Rate
(Lost Days × 1,000,000) ÷ Man-hrs
IOGP

DART

Days Away / Restricted / Transfer
((LTI + RWC) × 200,000) ÷ Man-hrs
OSHA

FAR

Fatal Accident Rate
(Fatalities × 100,000,000) ÷ Man-hrs
IOGP

NMR

Near Miss Ratio
Near Misses ÷ Recordable Injuries
ISO 45001

SAFE

Safe Days (LTI-Free)
Today - MAX(incident_date where severity=Lost Time)

CLO

Incident Close-Out Rate
Completed investigations ÷ Total × 100

MODULE 2 — RISK

ACR

CAPA Closure Rate
Completed CAPAs ÷ Total × 100
ISO 45001 §10

EFF

CAPA Effectiveness
AVG(effectiveness_rating) where Completed

OVR

Overdue CAPA Count
COUNT(status = Overdue)

MODULE 4 — ASSETS

PTW

PTW Compliance Rate
Properly Closed ÷ Total Issued × 100

LOTO

LOTO Compliance %
No-Deviation LOTO ÷ All LOTO × 100

location_station_id FK → working_stations

- ☐ **Controls (mitigations)**
controls Text
- ☐ **Register Status**
register_status Open / Under Control / Closed
- ☐ **GPS Location**
gps_latitude · gps_longitude **AUTO**
- ☐ **Review Notes**
review_notes Text **OPT**

Policy Management **policies**

- ☐ **Policy Name**
policy_name String(255) **REQ**
- ☐ **Category**
category String(100) — e.g. Safety / Environmental / HR
- ☐ **Issue Date**
issue_date Date
- ☐ **Owner**
owner String(100) — person or department name
- ☐ **Status**
status Draft / Active / Under Review / Archived

Auditor

HSE Auditor — PTW verification, hazard sign-off, compliance checklists, audit trail

Fields: 0/19

PTW Verification permits_to_work (update)

- ☐ Select Permit
id Select from open permits REQ
- ☐ Verification Result
verification_result Pass / Fail / Conditional REQ
- ☐ Verification Notes
verification_notes Text
- ☐ Auditor Verified By
auditor_verified_by FK → employees AUTO
- ☐ Verified At
auditor_verified_at DateTime AUTO

Hazard Verification hazards (update)

- ☐ Select Hazard
id Select from hazard register REQ
- ☐ Verification Notes
verification_notes Text
- ☐ Auditor Verified By
auditor_verified_by FK → employees AUTO
- ☐ Verified At
auditor_verified_at DateTime AUTO

Compliance Checklist Submission checklists (stub → future)

- ☐ Checklist Template
template_id Select from checklist templates REQ
- ☐ Site / Zone
site Select from sites
- ☐ Shift
shift Morning / Afternoon / Night
- ☐ Per-Item Response
items[].response Yes / No / N/A per checklist item REQ
- ☐ Per-Item Remarks
items[].remarks Text per item OPT
- ☐ Submission Status
status Draft → Submitted

Incident Close-Out Review incidents (read + validate)

- ☐ Review Investigation Completeness
investigation_status + five_why_analysis COMPUTED
- ☐ Validate Closure Notes
closure_notes Read-only review

VALUES AUDITOR GETS

PERMIT COMPLIANCE

PTW%

PTW Compliance Rate
 $Closed \div Issued \times 100$
permits_to_work · 79.5% current

LOTO

LOTO Compliance %
 $No-deviation\ PT004 \div Total\ PT004$
permits_to_work · 90.7% current

DEV

Permit Deviation Rate
 $deviation_reported=Yes \div Total \times 100$
7.4% current

VER

My Verifications Pending
auditor_verified_at IS NULL

CLOSE-OUT & COMPLIANCE

CLO

Incident Close-Out Rate
 $Completed\ investigations \div Total \times 100$
67.9% current

EFF

CAPA Effectiveness Avg
 $AVG(effectiveness_rating) = 3.81/5$

AUD

Audit Trail Log
Full action log across all tables
GET /audit-trail

CHECKLIST METRICS

CHK

Checklist Completion Rate
 $Submitted \div Total\ templates \times 100$

HZD

Hazards Verified / Total
auditor_verified_at IS NOT NULL $\div Total$

☐ **Confirm Lessons Communicated**
communicated_to_teams Verify = Yes

☐ **Audit Sign-Off Note**
verification_notes Text added by auditor